

Using States

A state is a defined set of values in the configuration of an object. It depends on the previous state. For example,

- ▶ HalfFull – We can say a glass is in a state called 'HalfFull' if it is being filled with a liquid and has reached half of its total capacity.
- ▶ HalfEmpty - We could also have a state called HalfEmpty which is the state that occurs when the amount of liquid drops to half of the glass's capacity.

Both states represent the same amount of liquid, but we consider them different. Likewise, states in a program represent not just values but may include how the current values were reached.

When a state changes a “transition” is said to occur. This is an opportunity to make changes or take actions that depend on the movement to the new state. For example, if we had a scene in the country where the state variable has two states “daylight” and “night”.

- ▶ When the state changes to “daylight” at this transition the sky would be made blue, the scenery green and the sun would appear in the sky.
- ▶ Then when the state changes to “night” at this transition the sky would be made dark, stars would be shown, the countryside would be darkened

Here is a simple QML program that shows the change of state in the above example. We have two rectangles, the top one is the 'sky' and the bottom one is the 'ground'. We will animate the change from daylight to night. There will be two states, but we only need to define one since 'daylight' will be the default state. We will just go to 'night' by clicking and holding the left mouse button down, releasing the mouse button will reverse the process

```

• import QtQuick 1.0
•
• Rectangle {
•     id: mainRectangle
•     width: 600
•     height: 400
•     color: "black"
•
•     Rectangle {
•         id: sky
•         width: 600
•         height: 200
•         y: 0

```